PHYSICS

SECTION-I (i)

1) Ra1 of mass 50 kg was moving with initial velocity of $20\hat{i} + 30\hat{j} + 40\hat{k}$. G1 applied a net external force on him and the value of force was $100\hat{i} + 200\hat{j} + 300\hat{k}$ for 10s. Assume Ra1 to offer no resistance, the angle made by his velocity vector after 10s with x-axis will be :-

(A) $\cos^{-1}\frac{7}{\sqrt{165}}$

(B) $\cos^{-1}\frac{10}{\sqrt{165}}$

(C) $\cos^{-1}\frac{4}{\sqrt{165}}$

(D) 0°

2)

The dimensional formula for ε_0 is :

(A) $M^{-2}L^{-3}I^{3}T^{4}$

(B) $M^{-1}L^{-3}I^{2}T^{4}$

(C) $M^{-1}L^{-3}I^{3}T^{2}$

(D) $M^{-2}L^{-3}I^{2}T^{2}$

3) A charged particle of mass m = 2 kg and charge 1μC is projected from a horizontal ground at an angle θ = 45° with speed 10 ms^{-1}. In space, a horizontal electric field E = 2×10^{7} NC^{-1} exists. The range of the projectile is

(A) 20m

(B) 60m

(C) 200m

(D) 180m

4) Two completely different planets X and Y are each composed of the same material. The planet Y has radius twice as large as that of X. The ratio of the gravitational acceleration g_y at the surface of Y to the gravitational acceleration g_x at the surface of X is given by :-

(A) $\frac{1}{8}$

(B) $\frac{1}{2}$

(C) 2

(D) $\frac{1}{4}$

SECTION-I (ii)

1) The value of $\left|\vec{A}+\vec{B}-\vec{C}+\vec{D}\right|$ can be zero if :—

(A) $\left|\vec{A}\right| = 5,\ \left|\vec{B}\right| = 3,\ \left|\vec{C}\right| = 4,\ \left|\vec{D}\right| = 13$

(B) $\left|\vec{A}\right| = 2\sqrt{2},\ \left|\vec{B}\right| = 2,\ \left|\vec{C}\right| = 2,\ \left|\vec{D}\right| = 5$

(C) $\left|\vec{A}\right| = 2\sqrt{2},\ \left|\vec{B}\right| = 2,\ \left|\vec{C}\right| = 2,\ \left|\vec{D}\right| = 10$

(D) $\left|\vec{A}\right| = 5,\ \left|\vec{B}\right| = 4,\ \left|\vec{C}\right| = 3,\ \left|\vec{D}\right| = 8$

2) The angle between two vectors $\vec{a}$ & $\vec{b}$ is θ and the magnitude of $\vec{b}$ is half of magnitude of $\vec{a}$. If $\vec{c} = \vec{a} - \vec{b}$ & $|\vec{a}| = a$ then choose the correct statements

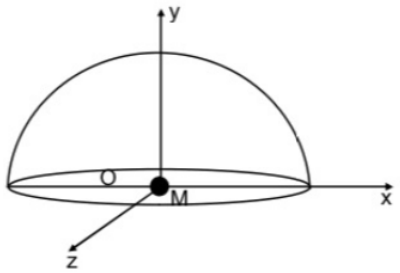
(A) if $c = \frac{a\sqrt{5}}{2}$ then θ will be 90°

(B) if $c = \frac{a\sqrt{3}}{2}$ then θ will be 60°

(C) if $c = \frac{a}{\sqrt{2}}$ then θ will be 45^{0}

(D) if $c = \frac{3a}{2}$ then θ will be 180°

3) A hemispherical shell of mass 2M and radius 6R and a point mass M are performing circular motion due to their mutual gravitational interaction. Their positions are shown in figure at any moment of time during motion. If r_1 and r_2 are the radii of circular path of hemispherical shell and point mass respectively and ω_1 and ω_2 are the angular speeds of hemi-spherical shell and point mass



respectively, then choose the correct statement(s).

(A) $\frac{r_1}{r_2} = \frac{2}{1}$

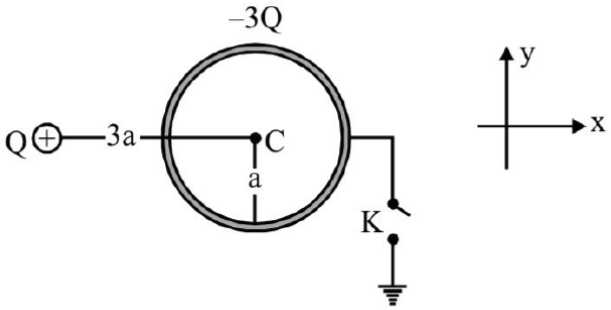
(B) $\frac{r_1}{r_2} = \frac{1}{2}$

(C) $\omega_1 = \sqrt{\frac{GM}{72R^3}}$

(D) $\omega_2 = \sqrt{\frac{GM}{72R^3}}$

4) A point charge +Q is kept at a distance 3a from the centre of a spherical conducting shell of charge -3Q and radius a.

Assuming $K = \frac{1}{4\pi\varepsilon_0}$, mark the **CORRECT** statements(s):-



Before closing switch the electric field at C due to the spherical shell is

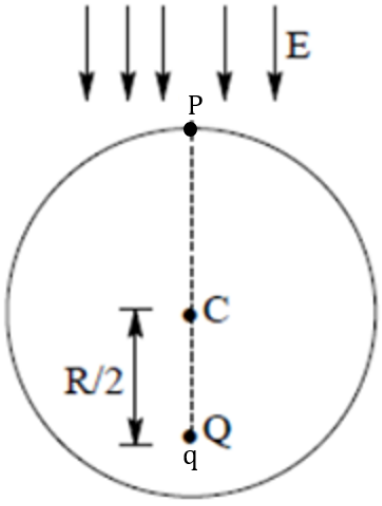
(A) $\left(-\frac{KQ}{9a^2}\hat{i}\right)$

(B) Before closing switch the potential of the shell is $\left(-\frac{8KQ}{3a}\right)$

(C) After closing the key,the charge on the shell is $\left(-\frac{Q}{3}\right)$

(D) After closing the key, the charge of the shell is zero

5) A positive charge q is placed inside a neutral hollow conducting sphere of radius R, as shown in figure. Whole system is placed in uniform external vertical electric field pointing downward (line PCQ is also vertical) then select the correct statement(s) about electric field at point P. Point P is a point of the material inside the conductor.



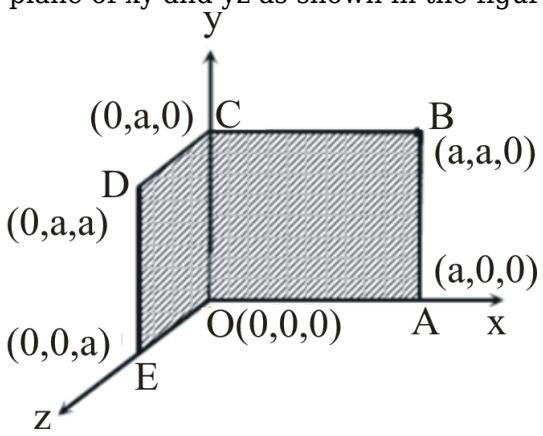
(A) Electric field due to outer surface of sphere at point P is zero

(B) Magnitude of electric field due to inner surface of sphere at point Pis $q/9\pi\varepsilon_0 R^2$

(C) Magnitude of electric field at point P, due to hollow sphere is less than E

(D) Direction of resultant electric field at point P due to charge q and hollow sphere is vertically upward

6) Two surface OABC and OCDE lies in the plane of xy and yz as shown in the figure. A charged



particle 'q' lies in the space at a point P. if

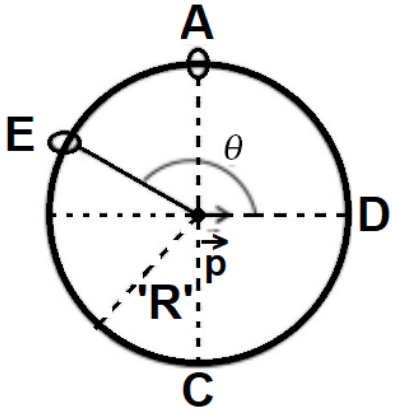
(A) coordinates of 'P' is (a − Δr, a − Δr, Δr) and a >> Δr, then flux passing through surface OABC is $\frac{7q}{24\varepsilon_0}$

(B) coordinates of 'P' is (a − Δr, a − Δr, Δr) and a >> Δr, then flux passing through surface OCDE is $\frac{q}{24\varepsilon_0}$

(C) coordinates of 'P' is (a + Δr, a + Δr, −Δr) and a >> Δr, then flux passing through surface OABC is $\frac{q}{24\varepsilon_0}$

(D) coordinates of 'P' is (a + Δr, a + Δr, −Δr) and a >> Δr, then flux passing through surface OCDE is $\frac{q}{24\varepsilon_0}$

7) A ring of radius R is fixed and kept in a vertical plane. A small dipole of dipole moment p is fixed at the centre of ring. A small bead of mass'm' and charge q is released from point A. Bead can slide on the ring. The inner diameter of bead and thickness of ring are nearly equal. Ring is insulating and frictionless. Ignore effect of gravity. Now choose the correct option(s).



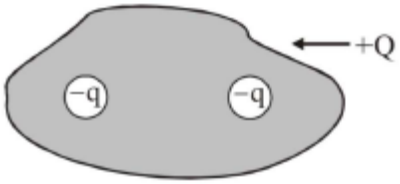
(A) Bead will never reach at point D

(B) The normal force between bead and ring at 'E' is directly proportional to the charge on the bead.

(C) The motion of bead is periodic.

(D) Speed of bead at point 'E' is given $\sqrt{\frac{-p(\cos\theta)q}{2\pi\varepsilon_0 mr^2}}$

8) A conducting body is given charge Q and charge -q has been placed in each of the cavity, which of the following statements is/are true?



(A) If Q = 2q, then conducting body will be at zero potential

(B) If an external electric field is applied then the charge distribution on the outer surface of conductor would change.

(C) The potential of any point inside the cavity is less than that of conducting body.

(D) None of these

SECTION-II

1) The maximum and minimum magnitudes of the resultant of two forces are 7 N and 1 N respectively. Find the magnitude of resultant force when they act perpendicular to each other :-

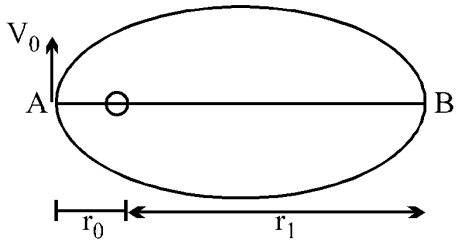
2) Two forces $\vec{F}_1$ and $\vec{F}_2$ have a resultant $\vec{F}_3$. If $\vec{F}_1 = 2\hat{i} - 3\hat{j}$ and $\vec{F}_3 = 5\hat{i} - 4\hat{j}$ then magnitude of $\vec{F}_2$ is X. The value of $\frac{X}{\sqrt{10}}$ is

3) Using dimensional analysis, angular frequency (ω) of surface waves on a lake as a function of surface tension σ, water density Q and the wave number k = 2π/λ (λ is the wavelength) is derived. Assuming that the dimensionless coefficient is equal to two (in SI units) in this case, find the value of group velocity v_g (in m/s) where $v_g = \frac{\partial\omega}{\partial k}$.
(Use : σ = 64mN·m^{-1}, Q = 320π kg·m^{-3}, λ = 1 cm)

4) Two particles of combined mass M, placed in space with certain separation, are released. Interaction between the particles is only of gravitational in nature and there is no external force present. Acceleration of one particle with respect to each other when separation between them is R, has a magnitude $\frac{xGM}{R^2}$. Value of x is

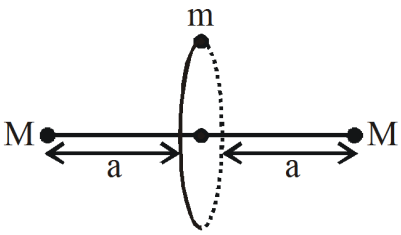
5) A satellite describes an elliptic orbit about a planet. Denoting by r_0 and r_1, the distances corresponding to the perigee and apogee of the orbit . The radius of curvature of the orbit at point A

0



and B in terms of r and r_1 is $\frac{\alpha r_0 r_1}{3(r_0 + r_1)}$. The value of α is.

6) A particle of mass m is moving in circular motion of radius r with constant speed v in influence of two fixed particles of mass M. Plane of circular motion is perpendicular to line joining fixed particles as shown. Speed of particle is given as $v = \sqrt{\frac{GM}{nr}}$, find value of n ($a = r\sqrt{3}$)



ANSWER KEYS

PHYSICS

SECTION-I (i)

Q.	1	2	3	4
A.	**C**	**B**	**A**	**C**

SECTION-I (ii)

Q.	5	6	7	8	9	10	11	12
A.	**B,D**	**A,B,D**	**B,C,D**	**A,B,C**	**B,D**	**A,B,C,D**	**A,B,C,D**	**A,B,C**

SECTION-II

Q.	13	14	15	16	17	18
A.	**5.00**	**1.00**	**0.60**	**1.00**	**6.00**	**4.00**

SOLUTIONS

PHYSICS

1)

$F_{net} = 100\hat{i} + 200\hat{j} + 300\hat{k} \Rightarrow \vec{a} = 2\hat{i} + 4\hat{j} + 6\hat{k}$

 velocity of Ra1 after 10 s = $40\hat{i} + 70\hat{j} + 100\hat{k}$

 Angle made by velocity vector with x-axis = $\cos^{-1}\frac{4}{10\sqrt{165}}$

2) $F = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2}$

$[\varepsilon_0] = \left[\frac{q_1 q_2}{4\pi F r^2}\right]$

$$a_x = \frac{qE}{m} = \frac{10^{-6}\times 2\times 10^7}{2} = 10 ms^{-2}$$

$$\text{Time of flight } T = \frac{2u\sin\theta}{g} = \frac{2\times 10\times\frac{1}{\sqrt{2}}}{10}$$

$$T = \sqrt{2} \text{ sec}$$

$$\text{Hence, range } R = u_x T + \frac{1}{2}a_x T^2$$

$$R = 10\cos 45^\circ \times T + \frac{1}{2}a_x T^2$$

$$= \frac{10}{\sqrt{2}}\sqrt{2} + \frac{1}{2}\times 10\times 2$$

3) $\Rightarrow R = 20m$

4) $g = \frac{GM}{R^2} = \frac{4}{3}G\pi\rho R$

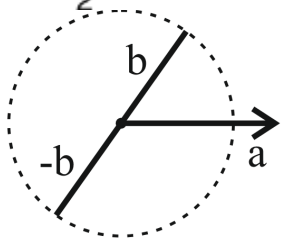
$\frac{g_y}{g_x} = \frac{R_y}{R_x} = \frac{8M_x}{4M_x} = 2$

5) $\left(\vec{A} + \vec{B} - \vec{C} + \vec{D}\right)$ can not be zero if sum of magnitude of three smallest vectors is less than the magnitude of bigger vector.

6) $\vec{a}.\vec{b} = \vec{a} + \left(-\vec{b}\right)$

$\Rightarrow \left|\vec{a}\right| = 2\left|\vec{b}\right| = a$

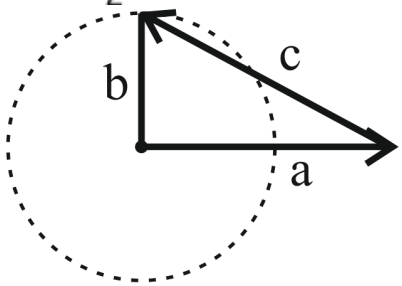
$|b| = \frac{a}{2}$



(A) $|\vec{c}| = \sqrt{|a|^2 + |b|^2 + 2\,|a|\,|b|\cos\theta}$

$\Rightarrow |\vec{c}| = \sqrt{a^2 + \frac{a^2}{4} + a \times a.\cos\theta}$

$|c| = \frac{\sqrt{5}a^2}{2}$



(B) $\frac{a\sqrt{3}}{2} = \sqrt{a^2 + \frac{a^2}{4} + a^2\cos\theta}$

$\Rightarrow \frac{3a^2}{2} = \frac{5a^2}{4} + a^2\cos\theta$

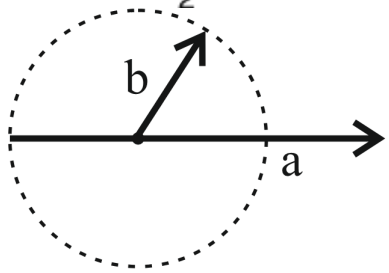
$\Rightarrow \frac{a^2}{4} = a^2\cos\theta$

$\Rightarrow \theta = 60^o$

(C) $\left(\frac{a}{\sqrt{2}}\right)^2 = \frac{5a^2}{4} + a^2\cos\theta$

$\Rightarrow \frac{a^2}{2} - \frac{5a^2}{4} = a^2\cos\theta$

$\Rightarrow \cos\theta = -\frac{\sqrt{3}}{2}$



$\left(\frac{3a}{2}\right)^2 = \frac{5a^2}{4} + a^2\cos\theta$

$\frac{9a^2}{4} - \frac{5a^2}{4} = a^2\cos\theta$

$\Rightarrow \cos\theta = -1$

$\theta = 180^0$

7) The system is equivalent to a binary system as shown in the diagram.

$r_1 = \frac{M \times 3R}{M + 2M} = R$

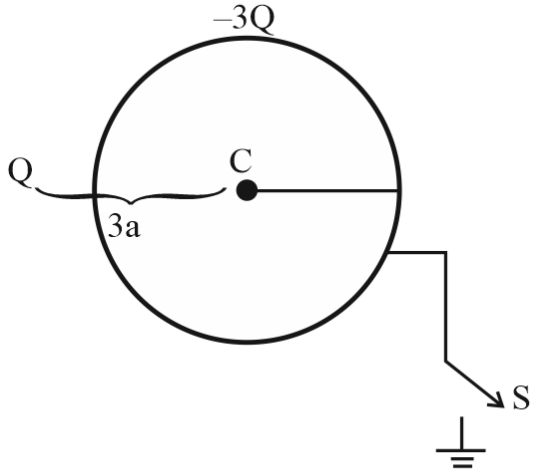
$r_2 = 2R$

Considering the circular motion of point mass m

$$M\omega_2^2 r_2 = \frac{GM^2}{36R^2}; \omega_2 = \sqrt{\frac{GM}{72R^3}} = \omega_1$$

8)

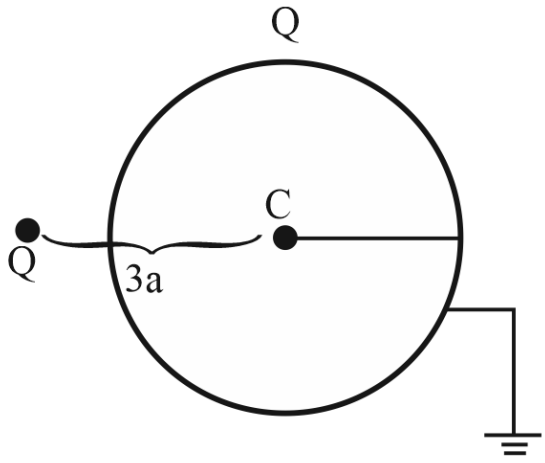
'S' i s open



$$\vec{E_C} = \vec{E}_{\text{due to O}} + \vec{E}_{\text{due to shell}}$$

$$\Rightarrow O = \frac{KQ}{9a^2}\hat{i} + \vec{E} \text{ due to shell}$$

$$V_C = \frac{KQ}{3a} + \frac{K(-3Q)}{a} = \frac{-8KQ}{3a}$$



$$V_C = O$$

$$\Rightarrow \frac{KQ}{3a} + \frac{Kq}{a} = 0$$

$$\Rightarrow q = -\frac{Q}{3}$$

'S' i s closed

9) External electric field and induced charge on outer surface of the sphere will produce a net zero field inside it. Net field at P must be vertically upward.

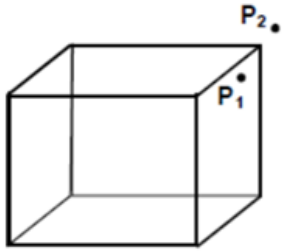
10) Using symmetry if charged particle lies at P_1 then $\phi_{OCDE} = \frac{q}{24\varepsilon_0}$

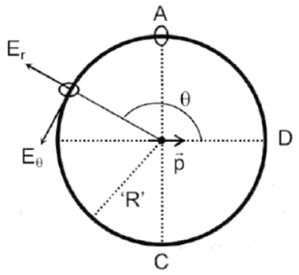
and $\phi_{OABC} = \frac{\left(\frac{q}{\varepsilon_0} - \frac{q}{8\varepsilon_0}\right)}{3} = \frac{7q}{24\varepsilon_0}$

If the charge particle lies at P_2 then

$\phi_{OCDE} = \frac{q}{24\varepsilon_0}$

and $\phi_{OABC} = \frac{q}{24\varepsilon_0}$





11)

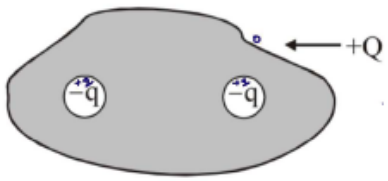
$E_r = \frac{2kp\cos\theta}{r^3}$

$E_\theta = \frac{kp\sin\theta}{r^3}$

From conservation of energy at A and E.

$0 + 0 = \frac{1}{2}mv^2 + \frac{(kp\cos\theta)q}{r^2}$

$v = \sqrt{\frac{-2kp(\cos\theta)q}{mr^2}} = \sqrt{\frac{-p(\cos\theta)q}{2\pi\varepsilon_0 mr^2}}$



12)

field outside zero potential decreases in the direction of field

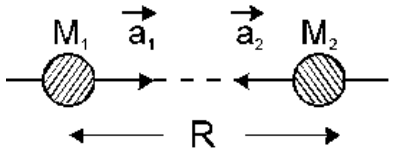
13) a + b = 7

a – b = 1

2a = 8
a = 4
b = 3
$R^2 = a^2 + b^2$
$R^2 = 4^2 + 3^2$
R = 5

14) $\vec{F}_1 + \vec{F}_2 = \vec{F}_3$
$\vec{F}_2 = \vec{F}_3 - \vec{F}_1$
$= (5\hat{i} - 4\hat{j}) - (2\hat{i} - 3\hat{j})$
$\vec{F}_2 = 3\hat{i} - \hat{j}$
$X = |\vec{F}_2| = \sqrt{10}$
$\frac{X}{\sqrt{10}} = 1$

15) Lets assume $\omega = 2\sigma^{\alpha}Q^{\beta}k^{\gamma}$.
Using dimensional analysis
$T^{-1} = (MT^{-2})\alpha\ (ML^{-3})^{\beta}\ (L^{-1})^{\gamma}$
On comparing
$0 = \alpha + \beta$,
$0 = -3\beta - \gamma$,
$-1 = -2\alpha$.
$\alpha = 1/2$, $\beta = -1/2$ and $\gamma = 3/2$
$\omega = 2\sqrt{\frac{\sigma k^3}{Q}}$
So, $v_g = \frac{\partial \omega}{\partial k} = 3\sqrt{\frac{\sigma k}{Q}}$
Substituting $k = 2\pi/\lambda$ and the rest of data given, we get
$v_g = 3\sqrt{\frac{2\pi\sigma}{Q\lambda}} = 0.6\ ms^{-1}$



16)
$a_1 = \frac{GM_1M_2}{R^2} / M_1$
$a_2 = \frac{GM_1M_2}{R^2} / M_2$
acceleation of M_1 w.r.t. M_2
$a_{rel.} = a_1 + a_2$
$= \frac{G(M_1 + M_2)}{R^2} = \frac{GM}{R^2}$

17) From property of ellipse

$a - ae = r_0$

$a + ae = r_1$

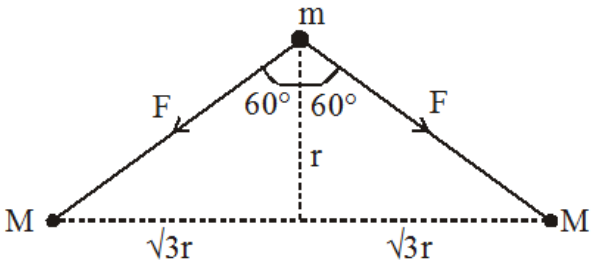
$$a = \frac{r_0 + r_1}{2}$$

$$e = \frac{r_1 - r_0}{r_1 + r_0}$$

Radius of curvature $= \frac{b^2}{a} = a^2 \times \frac{1 - e^2}{a} = \frac{2r_0 r_1}{r_0 + r_1}$

$\alpha = 6$

18)



$$2F\cos 60^\circ = 2\left[\frac{GMm}{(2r)^2}\right] \cdot \frac{1}{2} = \frac{mv^2}{r}$$

$$\Rightarrow V = \sqrt{\frac{GM}{4r}}$$